Smart Place: AI-Based Semantic Talent Matching Platform

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Overview

1. Introduction
2. Literature Survey
3. Proposed Method
4. Technologies Used

Introduction

Why This Project?

- Manual resume shortlisting takes a lot of time and effort.
- Keyword-based matching may reject good candidates due to missing exact words.
- Students need clear guidance on which jobs match their skills.
- Placement teams need an automated and accurate filtering method.
- Semantic matching helps understand the actual meaning of skills and requirements.

Literature Survey

Full Paper Title	**Complete List of Authors**	**Source**	**Key Finding**
Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks	Nils Reimers, Iryna Gurevych	Google Scholar	Optimized BERT for fast, accurate semantic text similarity.
A Systematic Review on Automated Resume Screening Systems	S. K. Ravindran, A. C. Mondal	IEEE	Verified that NLP automation reduces human bias and latency.
Information Extraction from Resumes Using Natural Language Processing	S. Tambe, S. Agrawal	Springer	Confirmed spaCy's efficiency in extracting skills from PDF layouts.

Literature Survey

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A Deep Learning Approach for Job-Resume Matching	L. Chen, R. Feng, B. Yin	IEEE	Semantic matching is more reliable than simple keyword frequency.
An Intelligent Recommender System for Recruitment	P. Gupta, S. Garg	Springer	Proposed ranking mechanisms to identify top candidates automatically.

Problem Statement

- Manual resume screening is time-consuming and inefficient.
- Keyword-based job matching fails to capture semantic meaning.
- Students do not receive personalized feedback on job suitability.
- Placement administrators spend significant time shortlisting candidates.
- Existing systems lack intelligent automation.
- An accurate and efficient matching system is required.

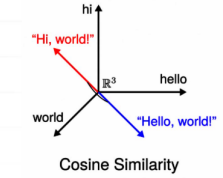
Proposed Solution

- Centralized web platform for student profiles and job postings
- Automatic resume analysis using NLP techniques
- Semantic comparison of student profiles and job descriptions
- Matching results displayed for student awareness
- Efficient candidate shortlisting for administrators

Technologies Used

Python

Pandas



Cosine Similarity

Streamlit

Sentence-BERT

SHA-256 Hashing

SQLite

spaCy

Project Flow

Student Profile / Resume Upload

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Resume Extraction + NLP Processing

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Job Description Input

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Semantic Matching (Embedding + Similarity)

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Match Score + Results Display

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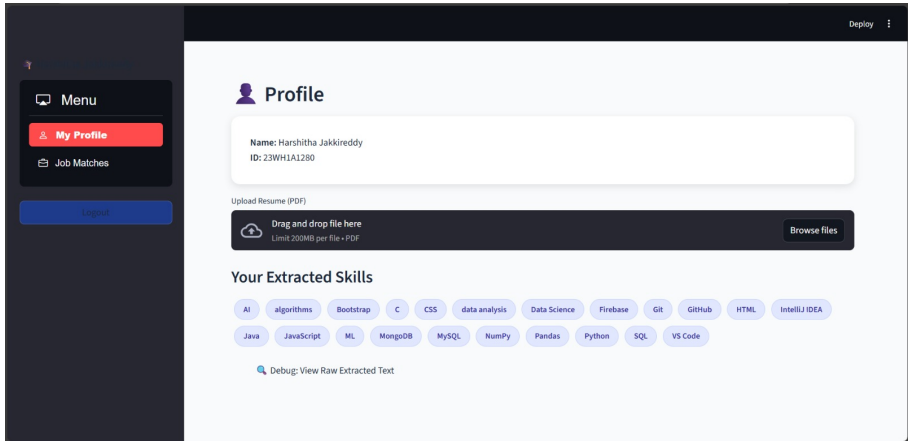
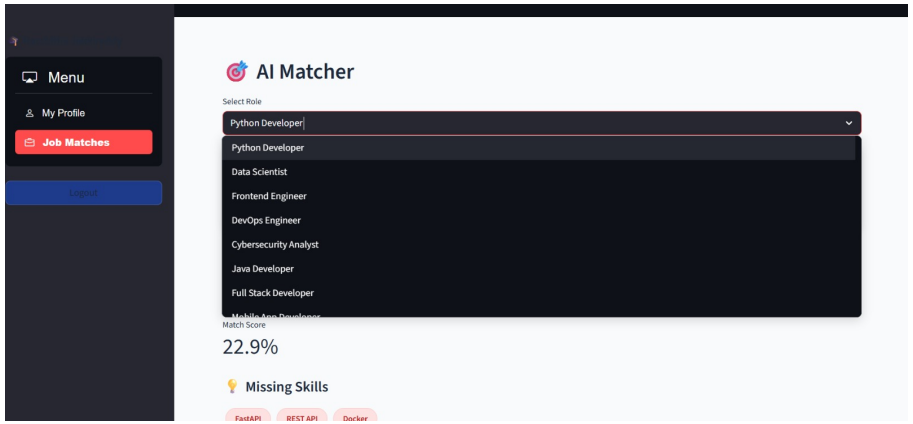
Shortlisting for Admin / Suggestions for Students

Implementation Status





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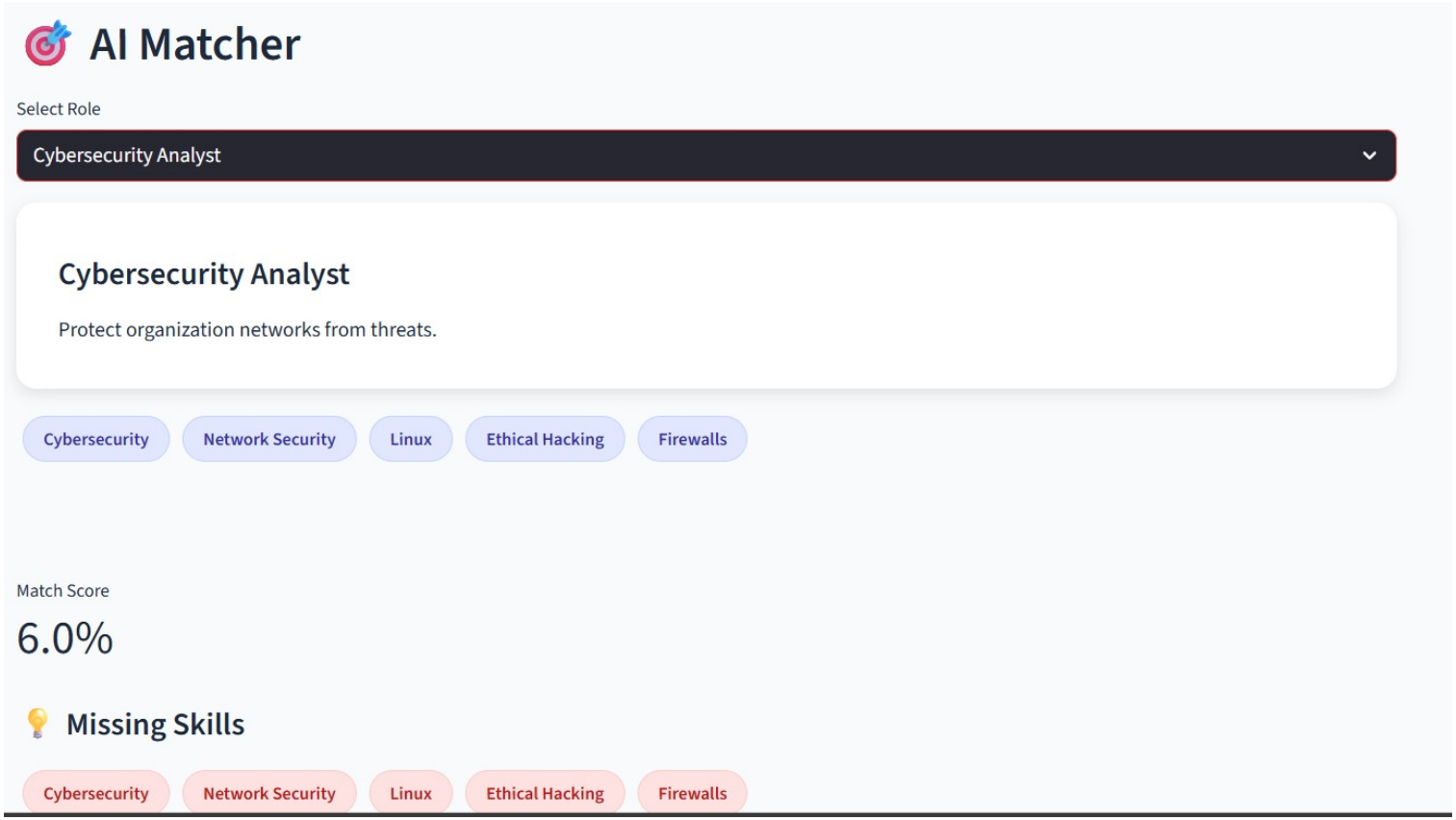

Thank you